SABARAGAMUWA UNIVERSITY OF SRI LANKA

FACULTY OF APPLIED SCIENCES

DEPARTMENT OF PHYSICAL SCIENCES AND TECHNOLOGY

Assignment Title:

Point of Sale (POS) System using C# Windows Forms and SQLite

Task:

Design and develop a desktop-based POS system for a small retail shop.

System Name:

AromaBliss POS (Retail Boutique: AromaBliss Studio)

Technologies:

C#, Windows Forms, SQLite

SUBMITTED BY:

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AromaBliss POS
Project Documentation & System Specification

1. **Project Overview & Business Domain**

- **Shop Name:** AromaBliss Studio (Artisanal Scents & Fragrances)
- **Business Domain:** A retail shop dealing in home fragrances, including scented candles, reed diffusers, pure essential oils, and room sprays.
- **System Purpose:** A standalone desktop Point of Sale (POS) application created using **C# Windows Forms** and **SQLite**. It helps store operators manage product inventory, track customer details, process checkout bills, update stock levels automatically, and inspect past sales history.
- **Intended Users:** Cashiers, sales operators, and shop owners.

2. System Login & User Access

To run and evaluate the application, use the following pre-configured login credentials:

Table 1: System Login Credentials & Access Control

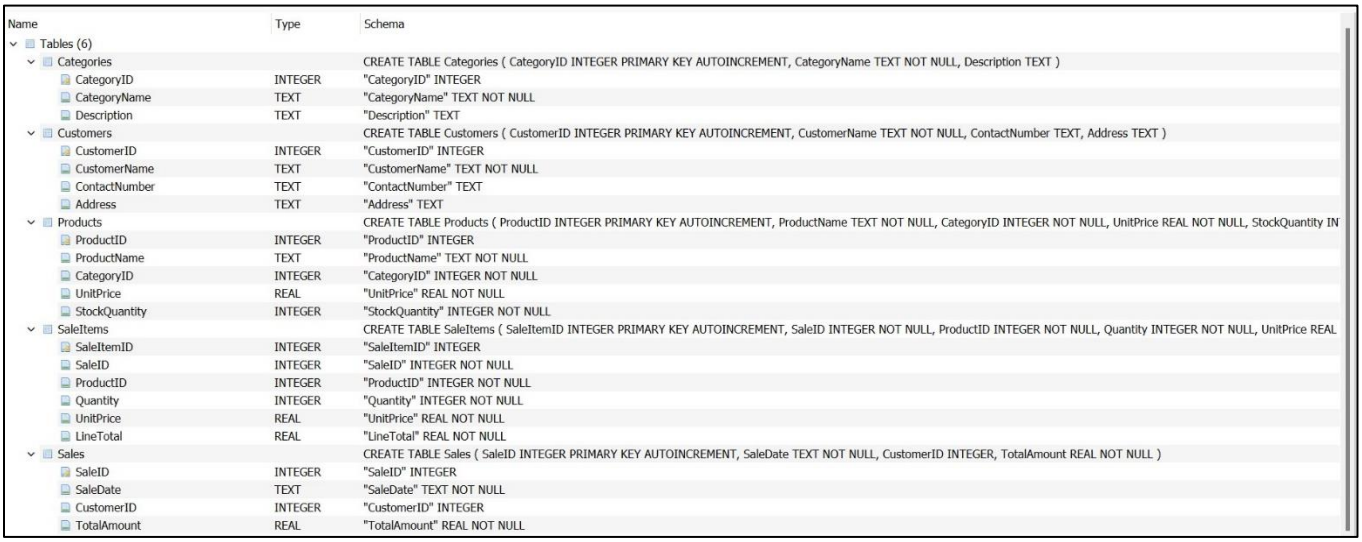
User Role	Username	Password	Access Level & Permissions
System Administrator	admin	1234	Full access to all features, including inventory, categories, customers, POS billing, and sales records.
Shop Owner	owner	1234	
Cashier	cashier	1234	
System Operator	operator	1234	

3. Database Architecture & Table Structure

The application connects to a local SQLite database (AromaBlissPOS.db) using a central DatabaseHelper.cs class.

Table 2: SQLite Database Table Structure & Data Types

Table Name	Columns & Data Types	Primary / Foreign Keys	Description
Categories	CategoryID *INTEGER*, CategoryName *TEXT*, Description *TEXT*	CategoryID (PK)	Stores fragrance product categories.
Products	ProductID *INTEGER*, ProductName *TEXT*, CategoryID *INTEGER*, UnitPrice *REAL*, StockQuantity *INTEGER*	ProductID (PK), CategoryID (FK)	Stores product details, selling prices, stock levels, and local image file paths.
Customers	CustomerID *INTEGER*, CustomerName *TEXT*, ContactNumber *TEXT*, Address *TEXT*	CustomerID (PK)	Stores customer contact details and delivery addresses.
Sales	SaleID *INTEGER*, SaleDate *TEXT*, CustomerID *INTEGER*, TotalAmount *REAL*	SaleID (PK), CustomerID (FK)	Stores completed transaction headers, dates, and total bill amounts.
SaleItems	SaleItemID *INTEGER*, SaleID *INTEGER*, ProductID *INTEGER*, Quantity *INTEGER*, UnitPrice *REAL*, LineTotal *REAL*	SaleItemID (PK), SaleID (FK), ProductID (FK)	Stores the itemized list of products, quantities, and line totals for each bill.



Name	Type	Schema
Tables (6)		
Categories		CREATE TABLE Categories (CategoryID INTEGER PRIMARY KEY AUTOINCREMENT, CategoryName TEXT NOT NULL, Description TEXT)
CategoryID	INTEGER	"CategoryID" INTEGER
CategoryName	TEXT	"CategoryName" TEXT NOT NULL
Description	TEXT	"Description" TEXT
Customers		CREATE TABLE Customers (CustomerID INTEGER PRIMARY KEY AUTOINCREMENT, CustomerName TEXT NOT NULL, ContactNumber TEXT, Address TEXT)
CustomerID	INTEGER	"CustomerID" INTEGER
CustomerName	TEXT	"CustomerName" TEXT NOT NULL
ContactNumber	TEXT	"ContactNumber" TEXT
Address	TEXT	"Address" TEXT
Products		CREATE TABLE Products (ProductID INTEGER PRIMARY KEY AUTOINCREMENT, ProductName TEXT NOT NULL, CategoryID INTEGER NOT NULL, UnitPrice REAL NOT NULL, StockQuantity IN
ProductID	INTEGER	"ProductID" INTEGER
ProductName	TEXT	"ProductName" TEXT NOT NULL
CategoryID	INTEGER	"CategoryID" INTEGER NOT NULL
UnitPrice	REAL	"UnitPrice" REAL NOT NULL
StockQuantity	INTEGER	"StockQuantity" INTEGER NOT NULL
SaleItems		CREATE TABLE SaleItems (SaleItemID INTEGER PRIMARY KEY AUTOINCREMENT, SaleID INTEGER NOT NULL, ProductID INTEGER NOT NULL, Quantity INTEGER NOT NULL, UnitPrice REAL
SaleItemID	INTEGER	"SaleItemID" INTEGER
SaleID	INTEGER	"SaleID" INTEGER NOT NULL
ProductID	INTEGER	"ProductID" INTEGER NOT NULL
Quantity	INTEGER	"Quantity" INTEGER NOT NULL
UnitPrice	REAL	"UnitPrice" REAL NOT NULL
LineTotal	REAL	"LineTotal" REAL NOT NULL
Sales		CREATE TABLE Sales (SaleID INTEGER PRIMARY KEY AUTOINCREMENT, SaleDate TEXT NOT NULL, CustomerID INTEGER, TotalAmount REAL NOT NULL)
SaleID	INTEGER	"SaleID" INTEGER
SaleDate	TEXT	"SaleDate" TEXT NOT NULL
CustomerID	INTEGER	"CustomerID" INTEGER
TotalAmount	REAL	"TotalAmount" REAL NOT NULL

Figure 1: SQLite Database Schema and Tables

4. System Modules & Form Specifications

4.1 Login Form (FrmLogin)

- **Purpose:** Acts as the entry screen to prevent unauthorized access to the application.
- **Key Features:** Username and password input fields with presence validation and direct routing to the main dashboard upon successful verification.

Table 3: Operator Authentication Form Controls

Field / Control	Control Type	Description & Validation
Username	TextBox	Required; enter admin/owner/cashier/operator.
Password	TextBox	Required; password-masked field, enter 1234.
Login	Button	Verifies login credentials and opens the dashboard.
Exit Application	Button	Safely closes the application.



Figure 2: Login Form

4.2 Main Dashboard (FrmDashboard)

- **Purpose:** Serves as the central menu, showing key store statistics and shortcuts to all main forms.
- **Key Features:** Overview metric cards, low-stock warning notifications, recent sales table, Light/Dark theme toggle, and direct navigation buttons.

Table 4: Central Dashboard Controls & Data Bindings

Component / Action	Type	Description / Data Source
Total Products Card	Panel /Label	Displays total count of active items from Products table.
Active Categories Card	Panel / Label	Displays total count of categories from Categories table.
Today's Sales Card	Panel / Label	Shows total revenue (LKR) generated during the current date.

Low Stock Alert	Label	Alerts the user when product quantities drop to 5 units.
Recent Transactions	DataGridView	Displays the last 5 completed sales (Invoice #, Date/Time, Customer ID, Total).
Theme Toggle	Button	Switches the user interface between Light and Dark themes.
Sidebar Menu	Button List	Navigates to Categories, Products, Customers, POS Billing, and Sales History.

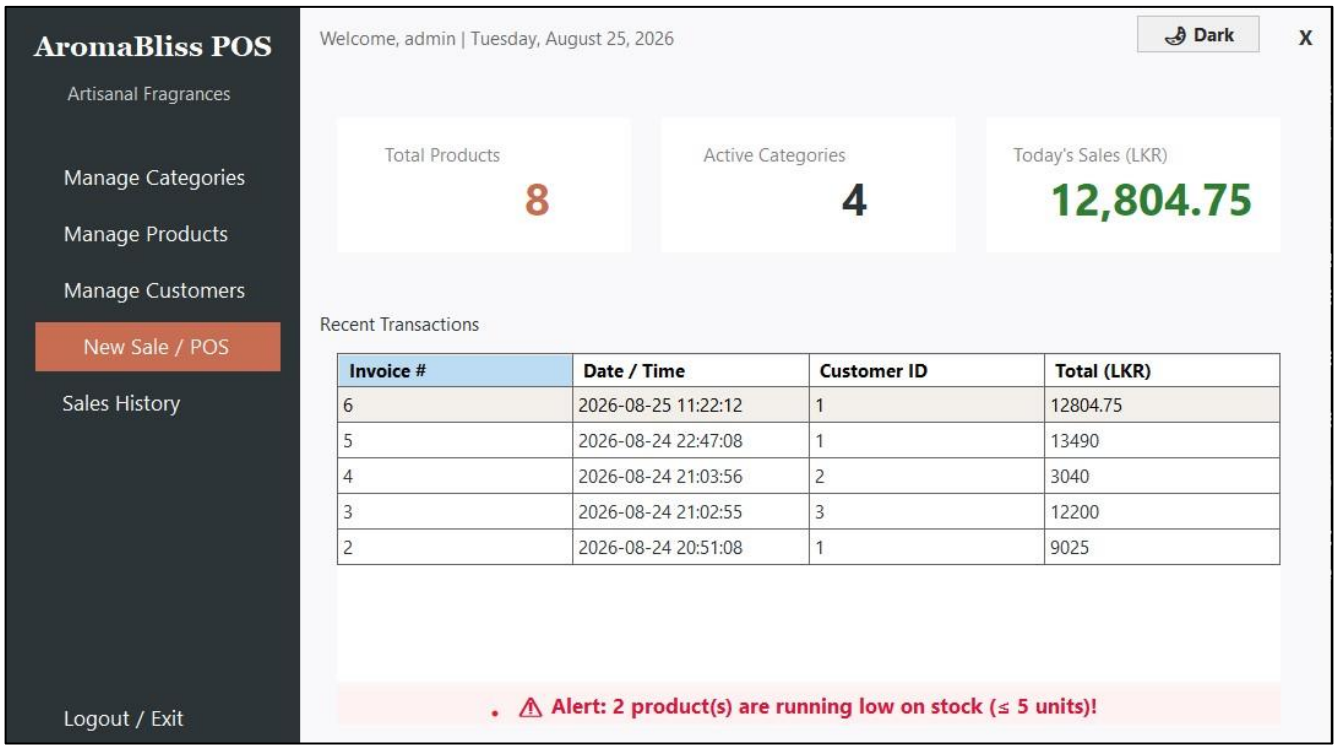


Figure 3: Main Dashboard Form (Light Mode)

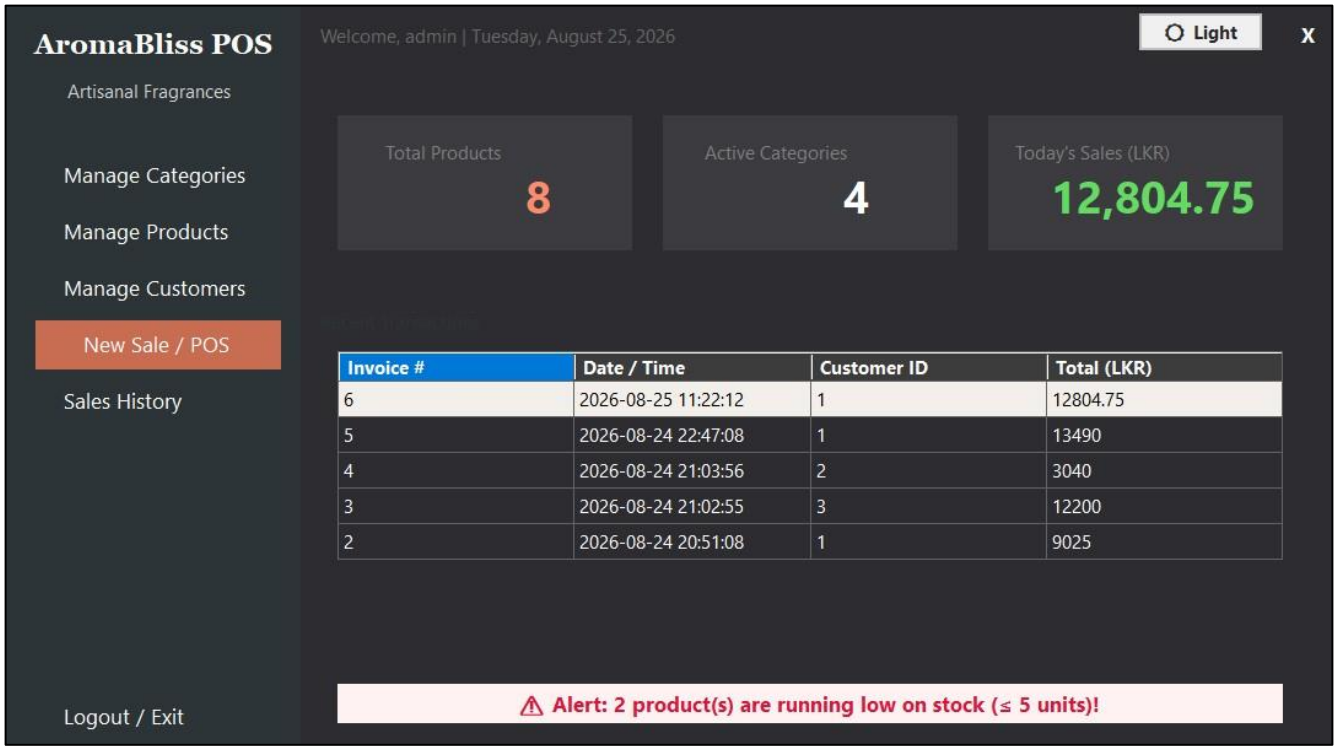


Figure 4: Main Dashboard Form (Dark Mode)

4.3 Category Management (FrmCategories)

- **Purpose:** Organizes the shop's fragrance collection into clear groups.
- **Key Features:** Full CRUD (Create, Read, Update, Delete) support with immediate table row binding.

Table 5: Category Management Form Controls & Validations

Field / Control	Control Type	Description & Validation
Category Name	TextBox	Required; name of category (e.g., Scented Candles, Reed Diffusers).
Description	TextBox	Optional; details or scent family description.
Save / Delete / Clear	Button	Action buttons to add, delete, or reset input fields.
Category List	DataGridView	Shows all saved categories (ID, Category, Description).

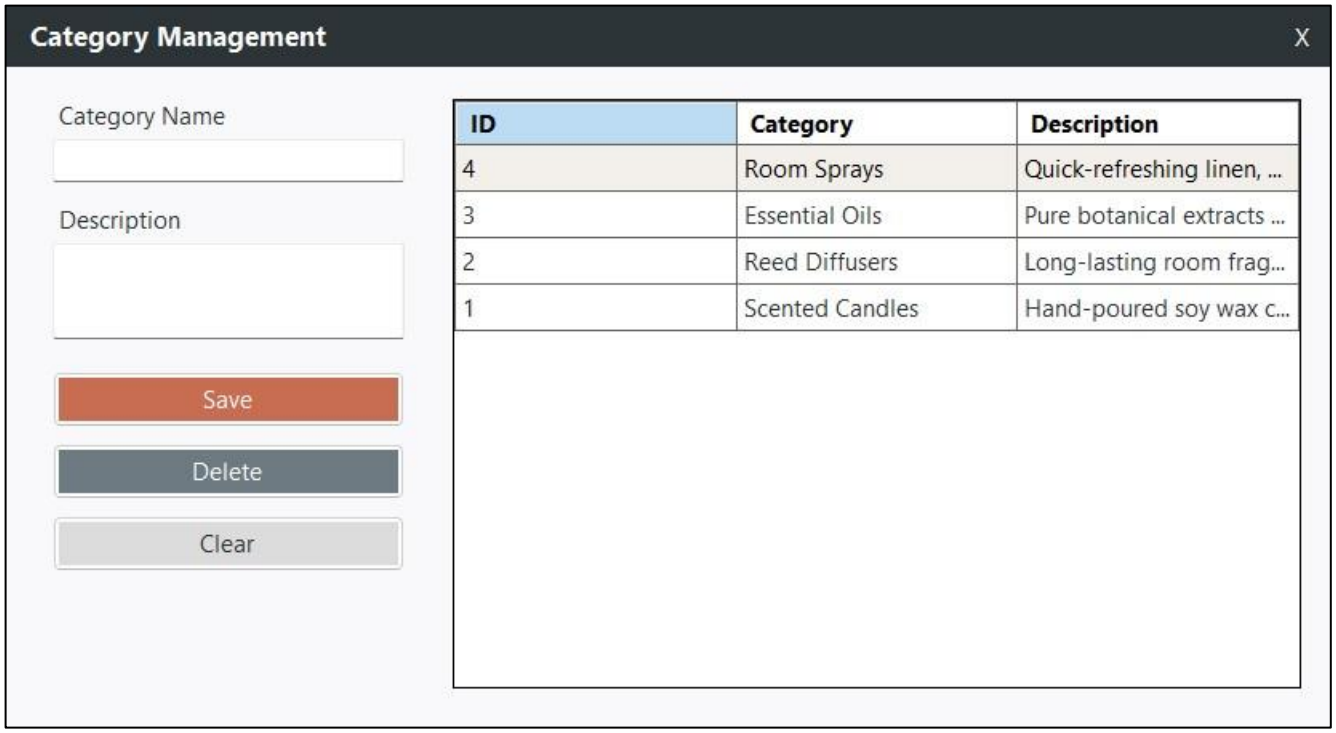


Figure 5: Category Management Form (Light Mode)

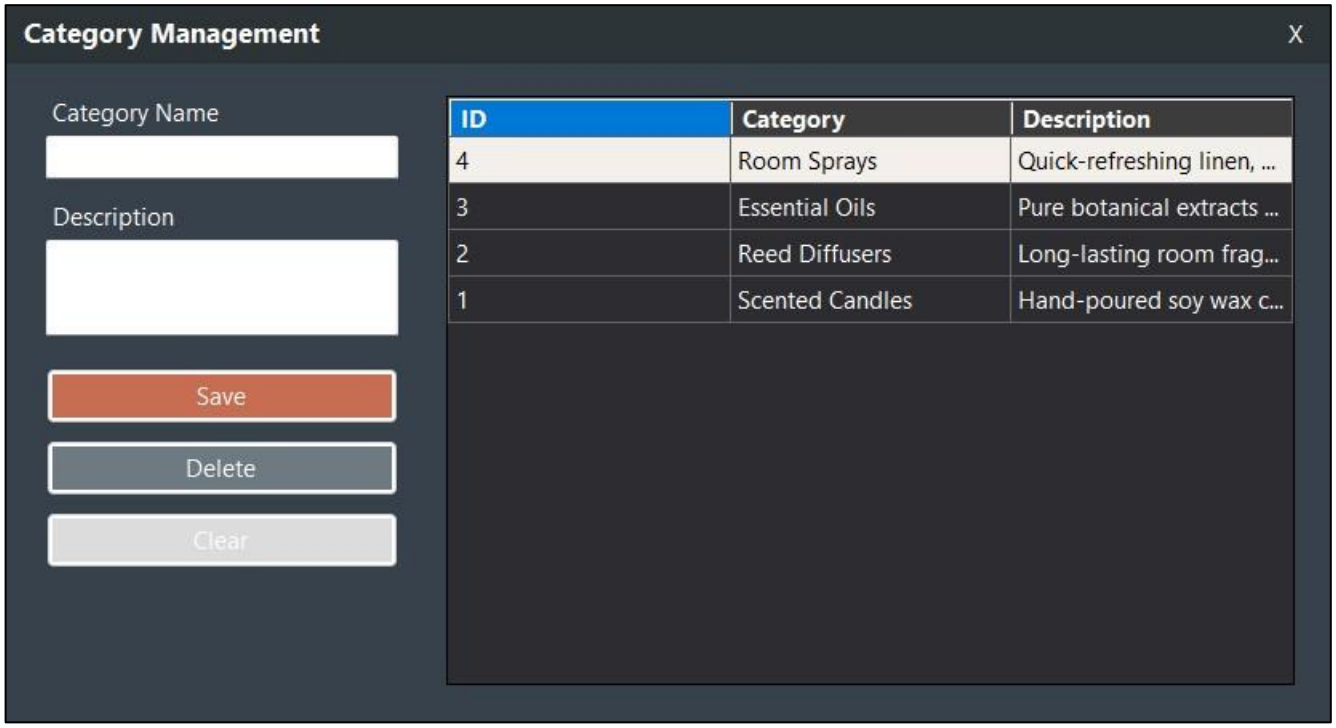


Figure 6: Category Management Form (Dark Mode)

4.4 Product Management (FrmProducts)

- **Purpose:** Handles the fragrance inventory, item pricing, stock numbers, and visual product pictures.
- **Key Features:** Live product search filter, large square image preview, and input validation rules.

Table 6: Product & Inventory Management Form Controls & Validations

Field / Control	Control Type	Description & Validation
Product Image	PictureBox	Displays a high-resolution preview of the product using Zoom mode.
Browse Image	Button	Opens OpenFileDialog to choose an image file from the computer.
Product Name	TextBox	Required; cannot be left blank.
Category	ComboBox	Required; dropdown list populated from Categories table.
Price (LKR)	TextBox	Numeric; must be greater than zero.
Stock Quantity	TextBox	Numeric; must be non-negative (>=0).
Search Product	TextBox	Live keyword filter to find products quickly.
Save / Delete / Clear	Button	Adds new products, updates existing items, or deletes entries.
Product Table	DataGridView	Displays ID, Product, Category, Price (LKR), and Stock.

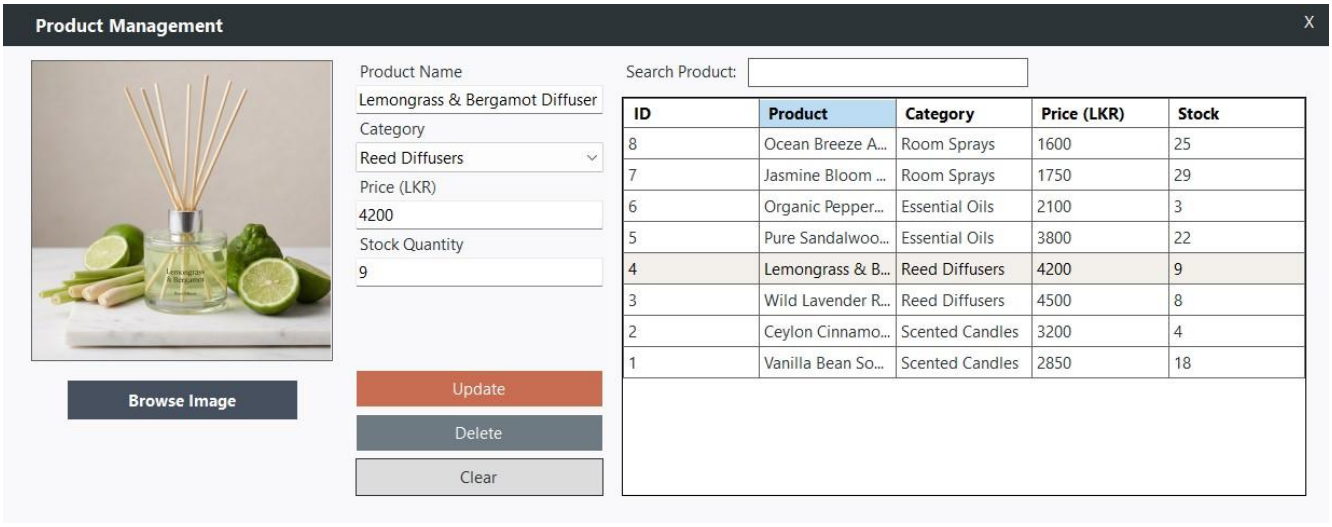


Figure 7: Product Management Form (Light Mode)

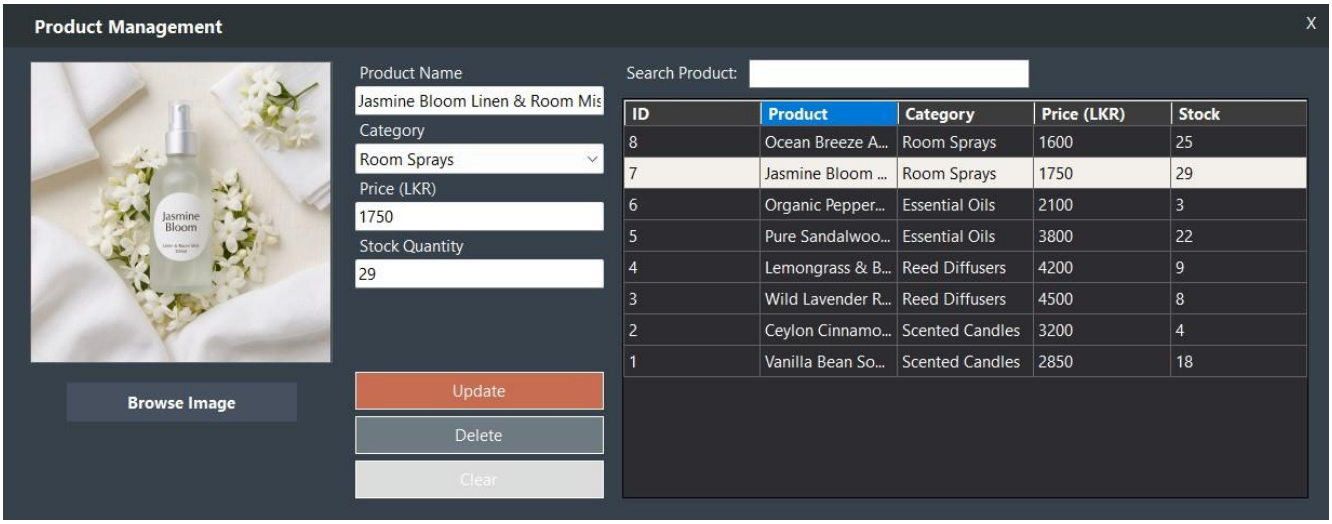


Figure 8: Product Management Form (Dark Mode)

4.5 Customer Management (FrmCustomers)

- **Purpose:** Stores contact records of customers for easy selection during billing.
- **Key Features:** Searchable customer table with contact numbers and addresses.

Table 7: Customer Management Form Controls & Validations

Field / Control	Control Type	Description & Validation
Customer Name	TextBox	Required; name of the customer.
Contact Number	TextBox	Contact phone number.
Address	TextBox	Customer location or delivery address.
Search Customer	TextBox	Live filter matching customer names.
Customer Table	DataGridView	Displays ID, Customer Name, Phone, and Address.

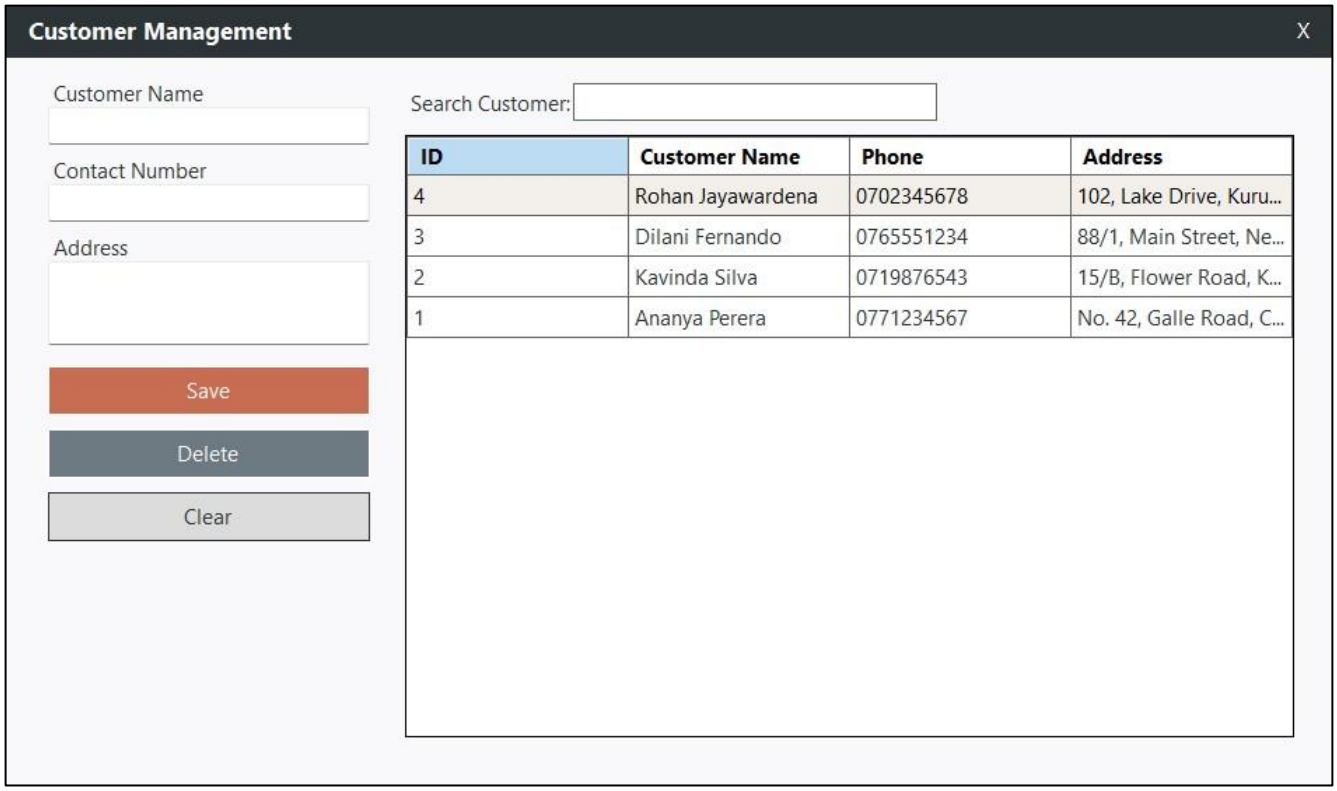


Figure 9: Customer Management Form (Light Mode)

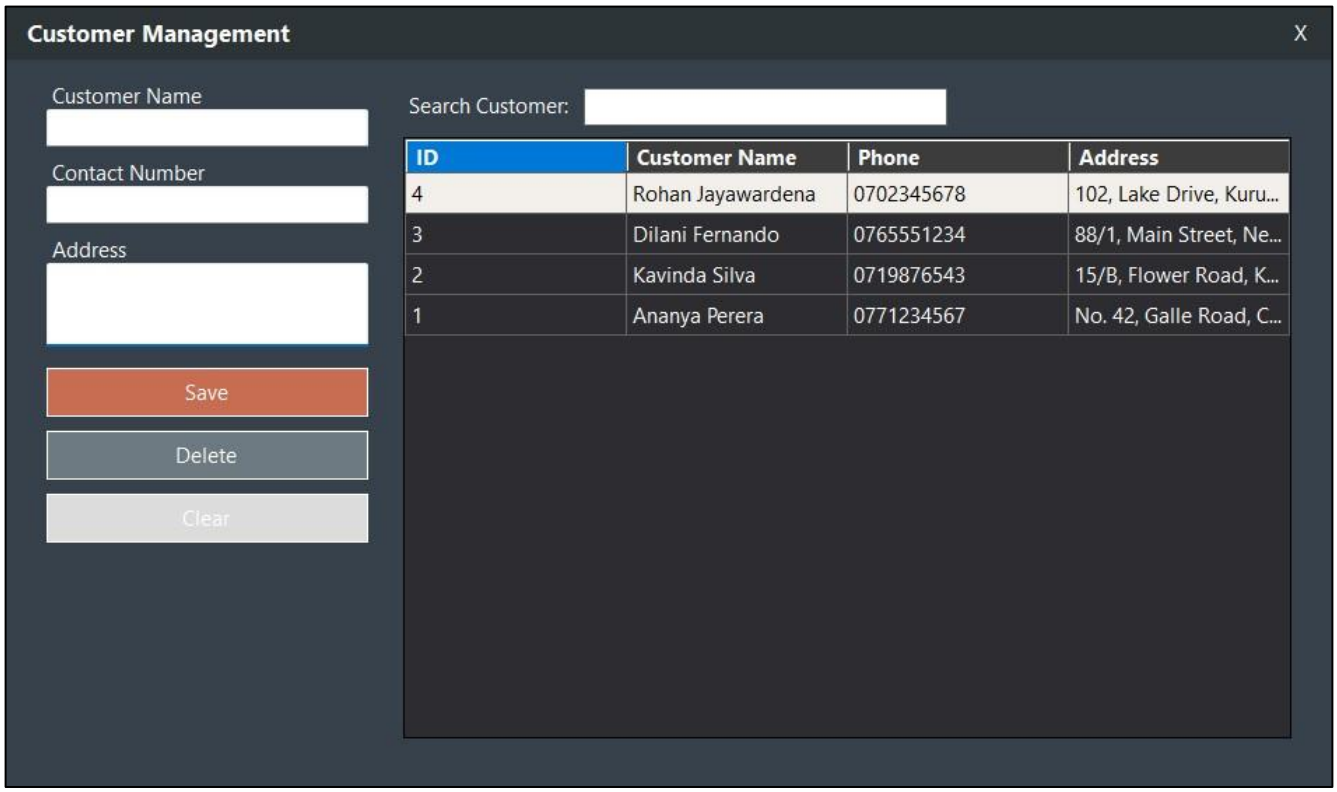


Figure 10: Customer Management Form (Dark Mode)

4.6 POS Billing Module (FrmPOS)

- **Purpose:** The main cash counter screen used to create orders and process customer sales.
- **Key Features:** Product dropdown selection, quantity stock checks, dynamic discount/tax calculation, automatic SQLite stock reduction, and receipt generation.

Table 8: POS Billing & Checkout Controls

Field / Control	Control Type	Description & Validation
Customer (Optional)	ComboBox	Selects a registered customer or leaves as Walk-in.
Select Product	ComboBox	Dropdown list of available store products.
Unit Price (LKR)	TextBox	Automatically loaded based on the selected item.
Quantity (Qty)	NumericUpDown	Must be >0 and cannot exceed current stock quantity.
Add / Remove Item	Button	Adds items to the cart or removes selected line items.
Cart Table	DataGridView	Displays Product, Unit Price, Qty, and Line Total.
Discount (%)	TextBox	Percentage discount applied to subtotal.
Tax (%)	TextBox	Percentage tax added to final amount.
Total Amount (LKR)	Label	Real-time calculated grand total.
Complete Sale	Button	Saves the bill into SQLite and decreases item stock numbers.
Print Receipt	Button	Opens the print preview window for the current bill.

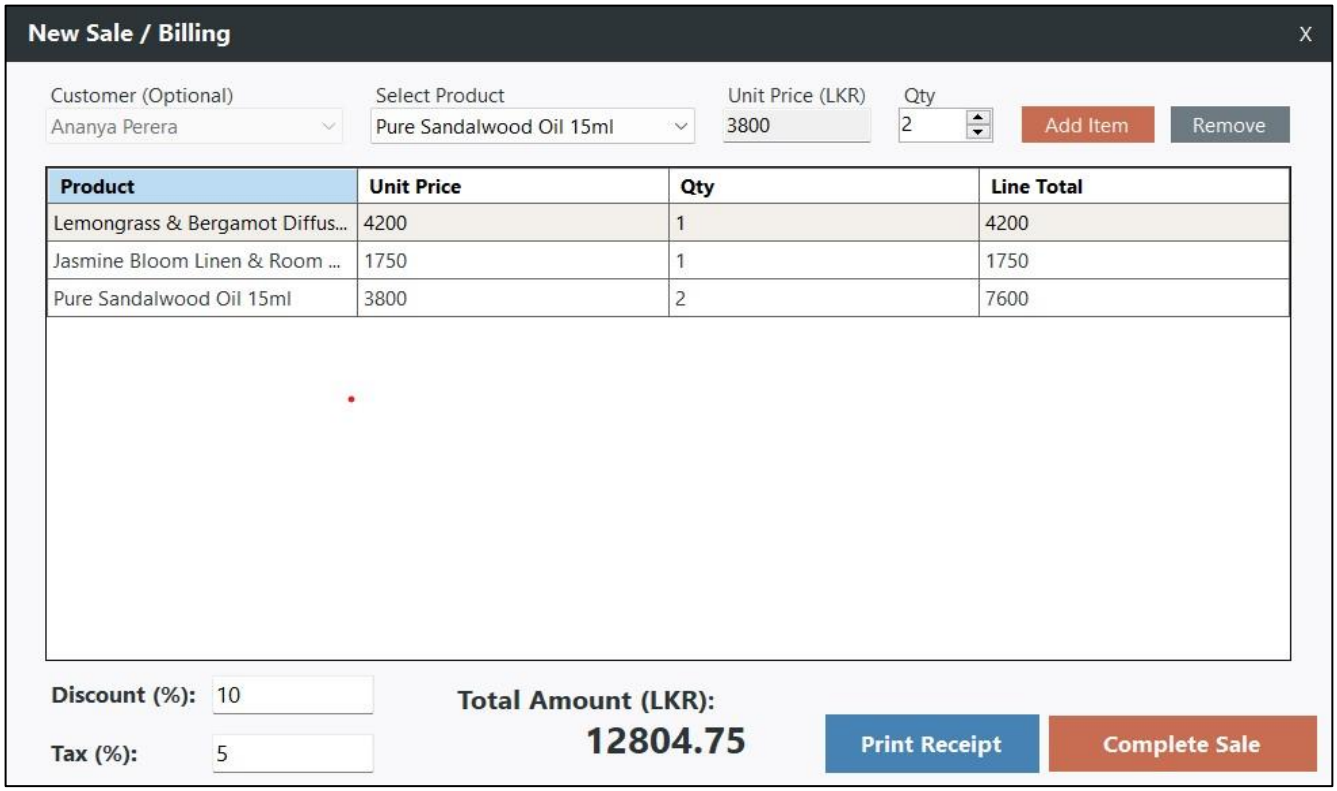


Figure 11: POS Billing Form (Light Mode)

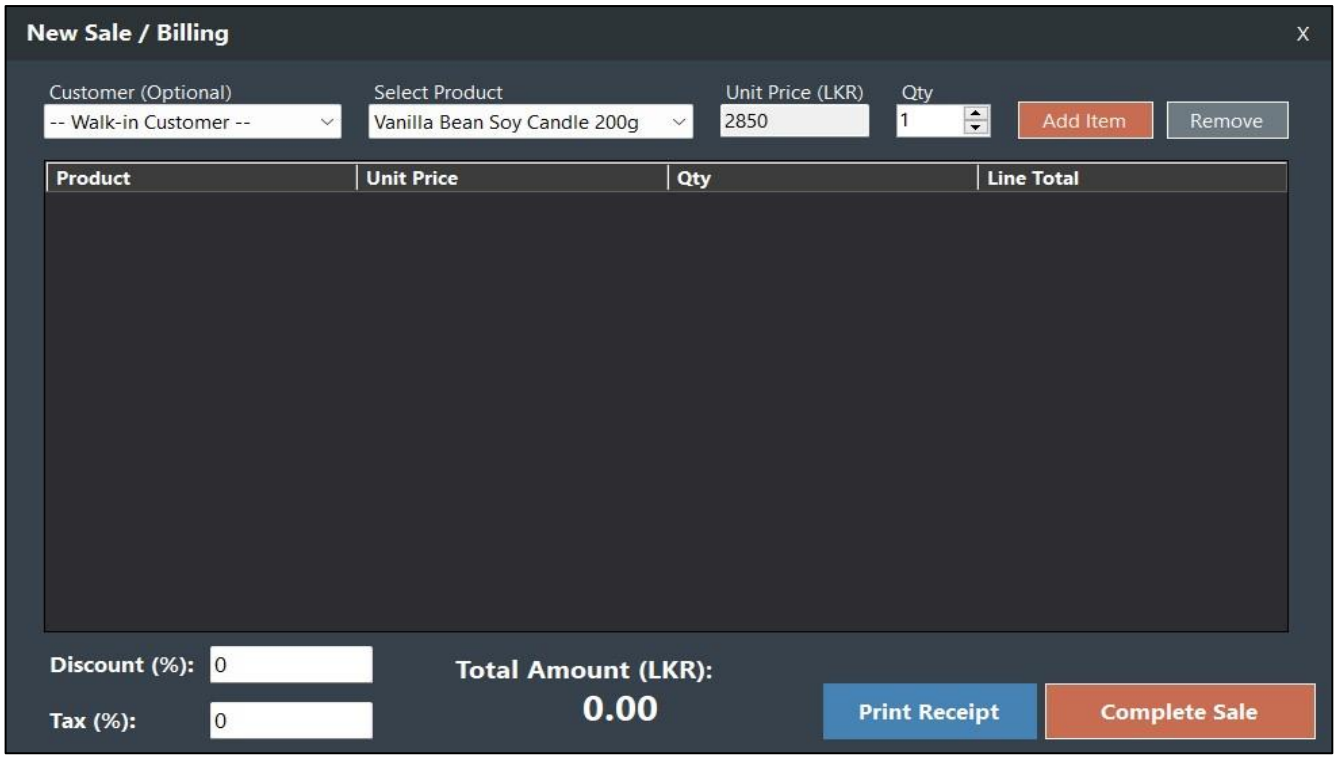


Figure 12: POS Billing Form (Dark Mode)

4.7 Receipt Print Preview

- **Purpose:** Displays a thermal-style printed receipt for the customer transaction.
- **Key Features:** Shows shop name, date/time stamp, cashier name, customer name, purchased items, discount/tax calculations, and total amount (LKR).

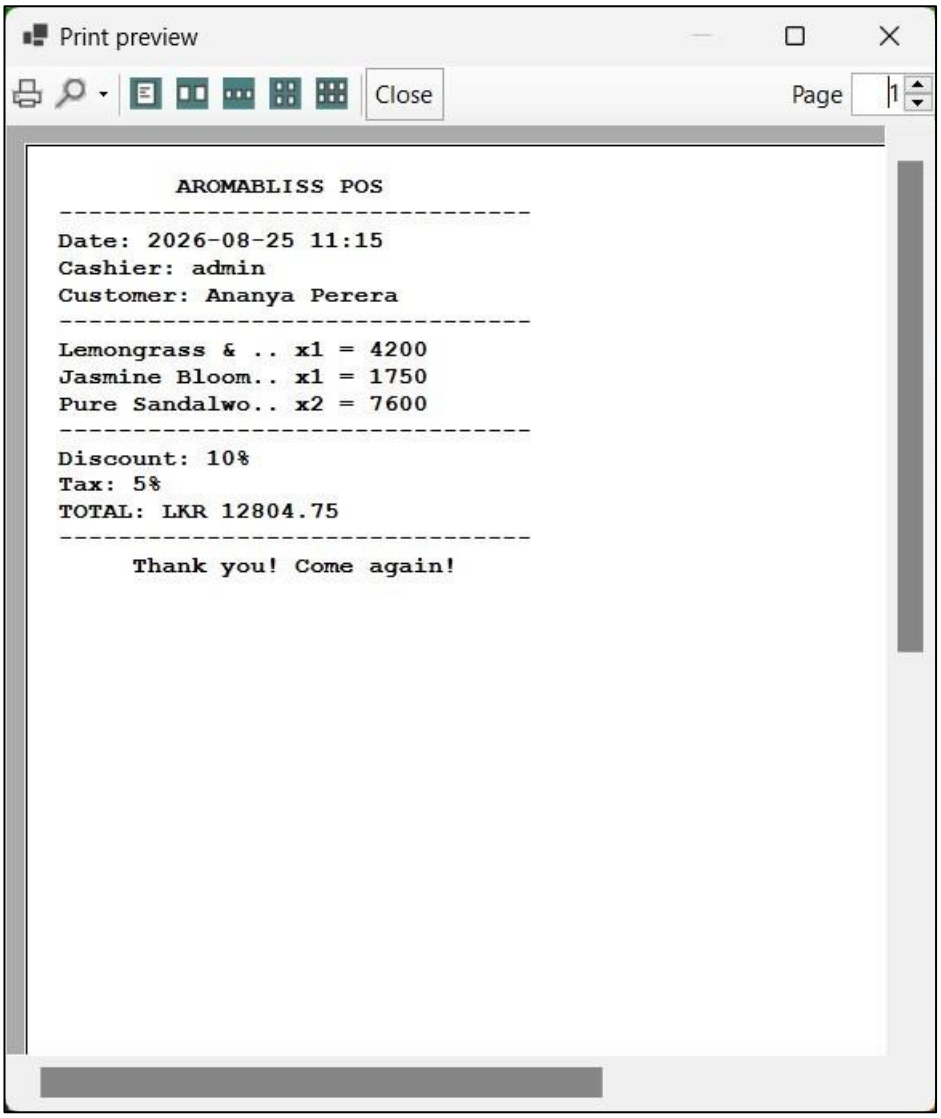


Figure 13: Thermal Receipt Print Preview

4.8 Sales History & Records (FrmSalesHistory)

- **Purpose:** Allows store operators to review and inspect previous transactions.
- **Key Features:** Search filter by invoice ID or customer name, and a master-detail view showing the receipt list and purchased line items.

Table 9: Sales History & Audit Controls

Field / Control	Control Type	Description & Functionality
Filter	TextBox	Search sales records by invoice number or customer name.
All Transactions	DataGridView	Displays Invoice #, Date, Customer, and Total (LKR).
Sale Item Breakdown	DataGridView	Displays Product, Qty, Price, and Total (LKR) for the selected bill.

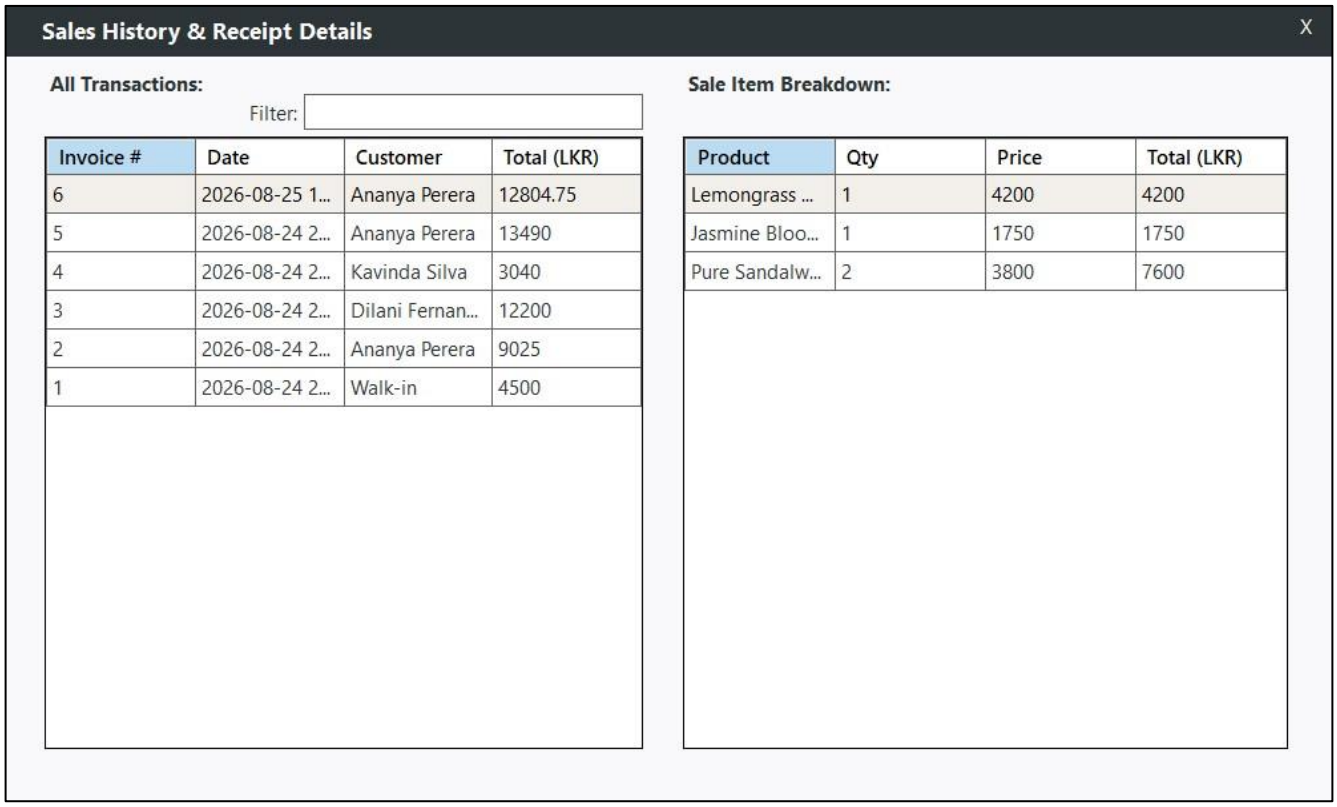


Figure 14: Sales History Form (Light Mode)

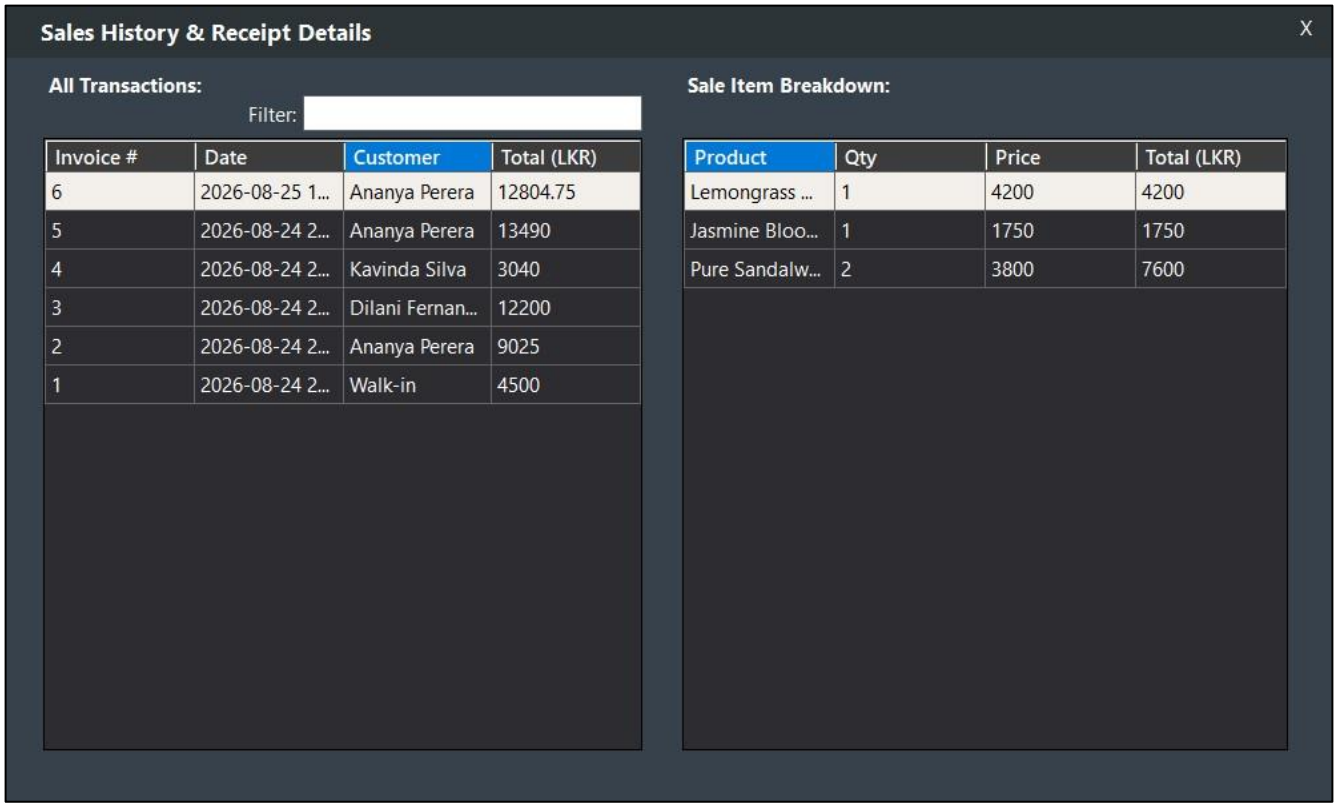


Figure 15: Sales History Form (Dark Mode)

4.9 Application Exit Confirmation

- **Purpose:** Confirms user intention before exiting the system to avoid accidental closure.
- **Key Features:** Displays a confirmation prompt box followed by a polite thank-you message.

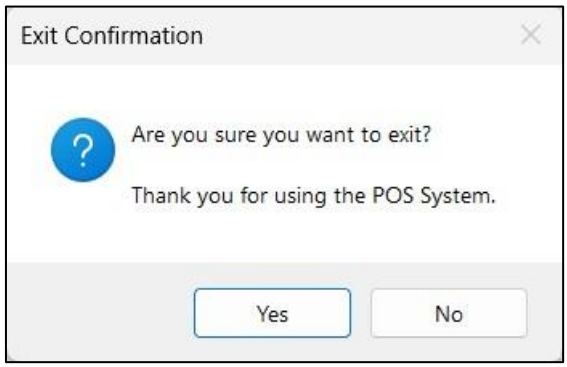


Figure 16: Exit Confirmation and Thank-You Message

5. System Features & Quality Checks

- **User Interface Design:** Includes clean control spacing, readable fonts, and full dual-theme support (Light Mode and Dark Mode).
- **Stock Synchronization:** Every completed sale in FrmPOS automatically updates the StockQuantity column in the SQLite database.
- **Data Validation:** Prevents common entry mistakes such as empty names, negative stock numbers, zero pricing, and selling more items than available in stock.